Statistics/Data analysis

```
      name:  <unnamed>
       log:  E:\Documents\STATA\birthweight_assignment2.smcl
  log type:  smcl
 opened on:   1 May 2026, 13:24:50

1 .
2 .
3 . *2)     A study was conducted to determine if smoking and mother's race are associat
  > ed with birthweight of babies. The data is attached as Regression assignment I Numbe
  > r_2_2025_2026 and the table below contains the details about the variables. Use it t
  > o answer the question. Interpret the final model obtained. USE STATA for the analysi
  > s
4 .
5 . * Import data
6 . import excel "E:\Documents\STATA\Regression assignment I data_Number 2_2025_2026.xls
  > x", sheet("Sheet1") firstrow
  (10 vars, 189 obs)

7 .
8 . * Rename columns
9 . rename Column1  id       // Identification code

10. rename Column2  age      // Age of mother in years

11. rename Column3  lwt      // Mother's weight at last menstrual period (lbs)

12. rename Column4  race     // Mother's race (1=White, 2=Black, 3=Other)

13. rename Column5  smoke    // Smoking during pregnancy (1=Yes, 0=No)

14. rename Column6  ptl      // History of premature labour

15. rename Column7  ht       // History of hypertension (1=Yes, 0=No)

16. rename Column8  ui       // Uterine irritability (1=Yes, 0=No)

17. rename Column9  ftv      // Physician visits - 1st trimester

18. rename Column10 bwt      // Birthweight in grams (OUTCOME)

19.
20. *check to see if renaming was succesful
21. bro

22.
23. *Label columns after renaming
24.
25. label variable id    "Identification Code"

26. label variable age   "Age of the Mother in Years"

27. label variable lwt   "Weight in Pounds at the Last Menstrual Period"

28. label variable race  "Mothers Race (1=White, 2=Black, 3=Other)"

29. label variable smoke "Smoking Status During Pregnancy (1=Yes, 0=No)"

30. label variable ptl   "History of Premature Labor (0=None, 1=One, etc.)"
```

```
31. label variable ht    "History of Hypertension (1=Yes, 0=No)"

32. label variable ui    "Presence of Uterine Irritability (1=Yes, 0=No)"

33. label variable ftv   "Number of Physician Visits During First Trimester"

34. label variable bwt   "Birth Weight in Grams"

35.
36. *Data inspection
37. bro

38.
39. *ptl and ftv have levels with sparse categories
40.
41. tab ptl
```

History of Premature Labor (0=None, 1=One, etc.)	Freq.	Percent	Cum.
0	**159**	**84.13**	**84.13**
1	**24**	**12.70**	**96.83**
2	**5**	**2.65**	**99.47**
3	**1**	**0.53**	**100.00**
Total	**189**	**100.00**	

```
42.
43. tab ftv
```

Number of Physician Visits During First Trimester	Freq.	Percent	Cum.
0	**100**	**52.91**	**52.91**
1	**47**	**24.87**	**77.78**
2	**30**	**15.87**	**93.65**
3	**7**	**3.70**	**97.35**
4	**4**	**2.12**	**99.47**
6	**1**	**0.53**	**100.00**
Total	**189**	**100.00**	

```
44.
45. *COLLAPSE SPARSE CATEGORIES;
46. *History of premature labor acategories(ptl); 0=159, 1=24, 2=5, 3=1 → Collapse 2 and
  >  3 into "2 or more"
47. gen ptl2 = ptl

48. recode ptl2 (2 3 = 2)
  (1 changes made to ptl2)

49. label define ptl2_lbl 0 "No prior premature labor"  1 "One prior premature labor"
  > 2 "Two or more prior premature labors"
```

```
50. label values ptl2 ptl2_lbl

51. tab ptl2
```

ptl2	Freq.	Percent	Cum.
No prior premature labor	159	84.13	84.13
One prior premature labor	24	12.70	96.83
Two or more prior premature labors	6	3.17	100.00
Total	189	100.00	

```
52.
53. * Number of Physician Visits During the First Trimester()ftv2: 0=100, 1=47, 2=30, 3=
  > 7, 4=4, 6=1 → Collapse 3, 4, 6 into "3 or more"
54. gen ftv2 = ftv

55. recode ftv2 (3 4 6 = 3)
  (5 changes made to ftv2)

56. label define ftv2_lbl 0 "No physician visits" 1 "One physician visit" 2 "Two physici
  > an visits" 3 "Three or more physician visits"

57.
58.
59. label values ftv2 ftv2_lbl

60. tab ftv2
```

ftv2	Freq.	Percent	Cum.
No physician visits	100	52.91	52.91
One physician visit	47	24.87	77.78
Two physician visits	30	15.87	93.65
Three or more physician visits	12	6.35	100.00
Total	189	100.00	

```
61.
62.
63. * SAMPLE SIZE CHECK
64. * We need atleast 40 subjects per variable
65. *
66. bro

67.
68. * FIT  MODEL
69. regress bwt smoke i.race age lwt i.ptl2 ht ui i.ftv2
```

Source	SS	df	MS		
Model	26789683.1	12	2232473.59	Number of obs	= 189
Residual	73127369.5	176	415496.418	F(12, 176)	= 5.37
				Prob > F	= 0.0000
Total	99917052.6	188	531473.684	R-squared	= 0.2681
				Adj R-squared	= 0.2182
				Root MSE	= 644.59

bwt	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
smoke	-293.3205	109.6431	-2.68	0.008	-509.7049	-76.93619
race						
2	-443.4656	149.8739	-2.96	0.004	-739.247	-147.6843
3	-300.2724	117.7204	-2.55	0.012	-532.5	

```
> 976
>       -67.94725
                                    |
                                age |  -2.106441   9.651227    -0.22   0.827    -21.15
> 347
>        16.94059
                                lwt |   4.458538   1.721434     2.59   0.010     1.061
> 229
>        7.855848
                                    |
                               ptl2 |
         One prior premature labor |  -320.2626    149.579    -2.14   0.034    -615.4
> 618
>       -25.06333
Two or more prior premature labors |   153.6952   277.8785     0.55   0.581    -394.7
> 076
>        702.0979
                                    |
                                 ht |  -575.0218    201.222    -2.86   0.005    -972.1
> 403
>       -177.9032
                                ui  |  -505.9775   137.8461    -3.67   0.000    -778.0
> 215
>       -233.9334
                                    |
                               ftv2 |
               One physician visit  |   118.7878   122.7752     0.97   0.335    -123.5
> 133
>        361.0888
            Two physician visits    |  -13.11499    139.511    -0.09   0.925    -288.4
> 447
>        262.2147
    Three or more physician visits  |  -112.8566    199.325    -0.57   0.572    -506.2
> 313
>        280.5181
                                    |
                              _cons |   2824.084   314.0448     8.99   0.000     2204.
> 305
>        3443.862
------------------------------------+----------------------------------------------------------------
  ----------------

70.
71. *CHECK ASSUMPTIONS
72.
73. * EXISTENCE OF A RELATIONSHIP
74.
75. correlate smoke race lwt ptl2 ht ui age ftv2 bwt
  (obs=189)

             |    smoke     race      lwt     ptl2       ht       ui      age     ftv2
>       bwt
-------------+------------------------------------------------------------------------
  ---------

       smoke |   1.0000
        race |  -0.3390   1.0000
         lwt |  -0.0442  -0.1650   1.0000
        ptl2 |   0.1837   0.0189  -0.1347   1.0000
          ht |   0.0134   0.0199   0.2364  -0.0133   1.0000
          ui |   0.0622   0.0536  -0.1528   0.2128  -0.1086   1.0000
         age |  -0.0443  -0.1728   0.1801   0.0717  -0.0158  -0.0752   1.0000
        ftv2 |  -0.0460  -0.1295   0.0884  -0.0512  -0.0711  -0.0505   0.2184   1.0000
         bwt |  -0.1891  -0.1962   0.1858  -0.1740  -0.1461  -0.2835   0.0899   0.0503
>    1.0000
```

```
76. *None of the predictor variables have a correlation coefficient > 0.3 with the birth
  > weight
77.
78. regress bwt smoke i.race age lwt i.ptl2 ht ui i.ftv2
```

Source	SS	df	MS		
Model	26789683.1	12	2232473.59	Number of obs	= 189
Residual	73127369.5	176	415496.418	F(12, 176)	= 5.37
				Prob > F	= 0.0000
				R-squared	= 0.2681
				Adj R-squared	= 0.2182
Total	99917052.6	188	531473.684	Root MSE	= 644.59

bwt	Coefficient	Std. err.	t	P>|t|	[95% conf. interval]
smoke	-293.3205	109.6431	-2.68	0.008	-509.7049 -76.93619
race					
2	-443.4656	149.8739	-2.96	0.004	-739.247 -147.6843
3	-300.2724	117.7204	-2.55	0.012	-532.5976 -67.94725
age	-2.106441	9.651227	-0.22	0.827	-21.15347 16.94059
lwt	4.458538	1.721434	2.59	0.010	1.061229 7.855848
ptl2					
One prior premature labor	-320.2626	149.579	-2.14	0.034	-615.4618 -25.06333
Two or more prior premature labors	153.6952	277.8785	0.55	0.581	-394.7076 702.0979
ht	-575.0218	201.222	-2.86	0.005	-972.1403 -177.9032
ui	-505.9775	137.8461	-3.67	0.000	-778.0215 -233.9334
ftv2					
One physician visit	118.7878	122.7752	0.97	0.335	-123.5133 361.0888
Two physician visits	-13.11499	139.511	-0.09	0.925	-288.4447 262.2147
Three or more physician visits	-112.8566	199.325	-0.57	0.572	-506.2313 280.5181
_cons	2824.084	314.0448	8.99	0.000	2204.305 3443.862

```
79. *However the F-test shows a p value <0.05; which shows there is existence of a relat
  > ionships between predictors and outcome; Therefore the assumption of existence of a
  > relationships is met
80.
81.
82. * CHECK FOR LINEARITY
83. * use CPR plots with each predictor.
84.
85. cprplot age, rlopts(lcolor(red)) lowess

86. *looking at this graph;  The CPR plot shows a straight reference line with no meanin
  > gful curvature, indicating that the linearity assumption is satisfied for age.the lo
  > wess line also closely aligns with with it for most of the data range; hence lineari
  > ty is fulfilled
87.
88. cprplot lwt, rlopts(lcolor(red)) lowess

89. *  straight reference line with no curvature, therefore linearity assumption is sati
  > sfied;lowess line here also closely aligns with the line of best fit therefore
90.
91. cprplot smoke, rlopts(lcolor(red)) lowess

92. * The line of best fit,along with the lowess smoother are straight indicating linear
  > ity
93.
94. regress bwt smoke race age lwt ptl2 ht ui ftv2
```

Source	SS	df	MS	Number of obs	=	189
				F(8, 180)	=	6.55
Model	22528099.7	8	2816012.47	Prob > F	=	0.0000
Residual	77388952.9	180	429938.627	R-squared	=	0.2255
				Adj R-squared	=	0.1910
Total	99917052.6	188	531473.684	Root MSE	=	655.7

bwt	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
smoke	-352.2265	107.4927	-3.28	0.001	-564.3345	-140.1186
race	-189.5989	57.89535	-3.27	0.001	-303.8398	-75.35806
age	.4082827	9.586487	0.04	0.966	-18.50807	19.32464
lwt	3.349241	1.687248	1.99	0.049	.0199116	6.67857
ptl2	-92.84892	108.0376	-0.86	0.391	-306.0321	120.3342
ht	-599.8101	203.596	-2.95	0.004	-1001.552	-198.0681
ui	-501.8176	139.5389	-3.60	0.000	-777.1602	-226.4751
ftv2	-26.56862	52.66522	-0.50	0.615	-130.4893	77.35201
_cons	3138.616	343.1829	9.15	0.000	2461.437	3815.795

```
95.
96. cprplot race

97. *race
98.
99.
100 regress bwt smoke i.race age lwt i.ptl2 ht ui i.ftv2
```

Source	SS	df	MS	Number of obs	=	189
				F(12, 176)	=	5.37
Model	26789683.1	12	2232473.59	Prob > F	=	0.0000
Residual	73127369.5	176	415496.418	R-squared	=	0.2681
				Adj R-squared	=	0.2182
Total	99917052.6	188	531473.684	Root MSE	=	644.59

bwt	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
smoke	-293.3205	109.6431	-2.68	0.008	-509.7049	-76.93619
race						
2	-443.4656	149.8739	-2.96	0.004	-739.2247	-147.6843
3	-300.2724	117.7204	-2.55	0.012	-532.5976	-67.94725
age	-2.106441	9.651227	-0.22	0.827	-21.15347	16.94059
lwt	4.458538	1.721434	2.59	0.010	1.061229	7.855848
ptl2						
One prior premature labor	-320.2626	149.579	-2.14	0.034	-615.4618	-25.06333
Two or more prior premature labors	153.6952	277.8785	0.55	0.581	-394.7076	702.0979
ht	-575.0218	201.222	-2.86	0.005	-972.1403	-177.9032
ui	-505.9775	137.8461	-3.67	0.000	-778.0215	-233.9334
ftv2						
One physician visit	118.7878	122.7752	0.97	0.335	-123.5133	361.0888
Two physician visits	-13.11499	139.511	-0.09	0.925	-288.4447	262.2147
Three or more physician visits	-112.8566	199.325	-0.57	0.572	-506.2313	280.5181
_cons	2824.084	314.0448	8.99	0.000	2204.305	3443.862

101
102

```
103 regress bwt smoke race age lwt ptl2 ht ui ftv2
```

Source	SS	df	MS		
Model	22528099.7	8	2816012.47	Number of obs	189
				F(8, 180)	6.55
Residual	77388952.9	180	429938.627	Prob > F	0.0000
				R-squared	0.2255
				Adj R-squared	0.1910
Total	99917052.6	188	531473.684	Root MSE	655.7

bwt	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
smoke	-352.2265	107.4927	-3.28	0.001	-564.3345	-140.1186
race	-189.5989	57.89535	-3.27	0.001	-303.8398	-75.35806
age	.4082827	9.586487	0.04	0.966	-18.50807	19.32464
lwt	3.349241	1.687248	1.99	0.049	.0199116	6.67857
ptl2	-92.84892	108.0376	-0.86	0.391	-306.0321	120.3342
ht	-599.8101	203.596	-2.95	0.004	-1001.552	-198.0681
ui	-501.8176	139.5389	-3.60	0.000	-777.1602	-226.4751
ftv2	-26.56862	52.66522	-0.50	0.615	-130.4893	77.35201
_cons	3138.616	343.1829	9.15	0.000	2461.437	3815.795

```
104
105 cprplot ptl2

106 *ptl2
107
108
109 regress bwt smoke i.race age lwt i.ptl2 ht ui i.ftv2
```

Source	SS	df	MS		
Model	26789683.1	12	2232473.59	Number of obs	189
				F(12, 176)	5.37
Residual	73127369.5	176	415496.418	Prob > F	0.0000
				R-squared	0.2681
				Adj R-squared	0.2182
Total	99917052.6	188	531473.684	Root MSE	644.59

bwt	Coefficient	Std. err.	t	P>|t|	[95% conf. interval]
smoke	-293.3205	109.6431	-2.68	0.008	-509.7049 -76.93619
race					
2	-443.4656	149.8739	-2.96	0.004	-739.247 -147.6843
3	-300.2724	117.7204	-2.55	0.012	-532.576 -67.94725
age	-2.106441	9.651227	-0.22	0.827	-21.15347 16.94059
lwt	4.458538	1.721434	2.59	0.010	1.061229 7.855848
ptl2					
One prior premature labor	-320.2626	149.579	-2.14	0.034	-615.4618 -25.06333
Two or more prior premature labors	153.6952	277.8785	0.55	0.581	-394.7076 702.0979

```
                                ht |  -575.0218    201.222    -2.86   0.005    -972.1
> 403
>       -177.9032
                                ui |  -505.9775   137.8461    -3.67   0.000    -778.0
> 215
>       -233.9334
                                   |
                              ftv2 |
                One physician visit |   118.7878   122.7752     0.97   0.335    -123.5
> 133
>        361.0888
             Two physician visits |  -13.11499    139.511    -0.09   0.925    -288.4
> 447
>        262.2147
      Three or more physician visits |  -112.8566    199.325    -0.57   0.572    -506.2
> 313
>        280.5181
                                   |
                              _cons |   2824.084   314.0448     8.99   0.000     2204.
> 305
>        3443.862
------------------------------------+----------------------------------------------------------------
```

```
110
111
112 *cprplot ptl2, rlopts(lcolor(red)) lowess
113 *The line of best fit is straight indicating linearity
114
115 cprplot ht

116 *The line of best fit is straight indicating linearity
117
118 cprplot ui

119 *The line of best fit is straight indicating linearity with no curvatures
120
121 regress bwt smoke race age lwt ptl2 ht ui ftv2
```

Source	SS	df	MS		
Model	22528099.7	8	2816012.47	Number of obs	= 189
Residual	77388952.9	180	429938.627	F(8, 180)	= 6.55
				Prob > F	= 0.0000
Total	99917052.6	188	531473.684	R-squared	= 0.2255
				Adj R-squared	= 0.1910
				Root MSE	= 655.7

bwt	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
smoke	-352.2265	107.4927	-3.28	0.001	-564.3345	-140.1186
race	-189.5989	57.89535	-3.27	0.001	-303.8398	-75.35806
age	.4082827	9.586487	0.04	0.966	-18.50807	19.32464
lwt	3.349241	1.687248	1.99	0.049	.0199116	6.67857
ptl2	-92.84892	108.0376	-0.86	0.391	-306.0321	120.3342
ht	-599.8101	203.596	-2.95	0.004	-1001.552	-198.0681
ui	-501.8176	139.5389	-3.60	0.000	-777.1602	-226.4751
ftv2	-26.56862	52.66522	-0.50	0.615	-130.4893	77.35201
_cons	3138.616	343.1829	9.15	0.000	2461.437	3815.795

```
122
123 cprplot ftv2

124 *ftv2
125
126
127 regress bwt smoke i.race age lwt i.ptl2 ht ui i.ftv2
```

Source	SS	df	MS		
Model	**26789683.1**	**12**	**2232473.59**	Number of obs	= **189**
				F(12, 176)	= **5.37**
Residual	**73127369.5**	**176**	**415496.418**	Prob > F	= **0.0000**
				R-squared	= **0.2681**
				Adj R-squared	= **0.2182**
Total	**99917052.6**	**188**	**531473.684**	Root MSE	= **644.59**

bwt	Coefficient	Std. err.	t	P>|t|	[95% conf. interval]
smoke	**-293.3205**	**109.6431**	**-2.68**	**0.008**	**-509.7049 -76.93619**
race					
2	**-443.4656**	**149.8739**	**-2.96**	**0.004**	**-739.247 -147.6843**
3	**-300.2724**	**117.7204**	**-2.55**	**0.012**	**-532.5976 -67.94725**
age	**-2.106441**	**9.651227**	**-0.22**	**0.827**	**-21.15347 16.94059**
lwt	**4.458538**	**1.721434**	**2.59**	**0.010**	**1.061229 7.855848**
ptl2					
One prior premature labor	**-320.2626**	**149.579**	**-2.14**	**0.034**	**-615.4618 -25.06333**
Two or more prior premature labors	**153.6952**	**277.8785**	**0.55**	**0.581**	**-394.7076 702.0979**
ht	**-575.0218**	**201.222**	**-2.86**	**0.005**	**-972.1403 -177.9032**
ui	**-505.9775**	**137.8461**	**-3.67**	**0.000**	**-778.0215 -233.9334**
ftv2					
One physician visit	**118.7878**	**122.7752**	**0.97**	**0.335**	**-123.5133 361.0888**
Two physician visits	**-13.11499**	**139.511**	**-0.09**	**0.925**	**-288.8447 262.2147**
Three or more physician visits	**-112.8566**	**199.325**	**-0.57**	**0.572**	**-506.2313 280.5181**
_cons	**2824.084**	**314.0448**	**8.99**	**0.000**	**2204.305 3443.862**

```
128
129 * CHECK FOR INDEPENDENCE OF RESIDUALS
130
131 * Use RVP plots (residuals versus predictor plots) - should show no clear pattern.
132
133 rvpplot age, yline(0)

134 *no pattern or funnel-shape noted
135
136 rvpplot lwt, yline(0)

137 *no pattern or funnel shaped
138
139 regress bwt smoke race age lwt ptl2 ht ui ftv2
```

Source	SS	df	MS		
Model	22528099.7	8	2816012.47	Number of obs	= 189
Residual	77388952.9	180	429938.627	F(8, 180)	= 6.55
				Prob > F	= 0.0000
				R-squared	= 0.2255
				Adj R-squared	= 0.1910
Total	99917052.6	188	531473.684	Root MSE	= 655.7

bwt	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
smoke	-352.2265	107.4927	-3.28	0.001	-564.3345	-140.1186
race	-189.5989	57.89535	-3.27	0.001	-303.8398	-75.35806
age	.4082827	9.586487	0.04	0.966	-18.50807	19.32464
lwt	3.349241	1.687248	1.99	0.049	.0199116	6.67857
ptl2	-92.84892	108.0376	-0.86	0.391	-306.0321	120.3342
ht	-599.8101	203.596	-2.95	0.004	-1001.552	-198.0681
ui	-501.8176	139.5389	-3.60	0.000	-777.1602	-226.4751
ftv2	-26.56862	52.66522	-0.50	0.615	-130.4893	77.35201
_cons	3138.616	343.1829	9.15	0.000	2461.437	3815.795

```
140
141 rvpplot race, yline(0)

142 *race
143 regress bwt smoke i.race age lwt i.ptl2 ht ui i.ftv2
```

Source	SS	df	MS		
Model	26789683.1	12	2232473.59	Number of obs	= 189
Residual	73127369.5	176	415496.418	F(12, 176)	= 5.37
				Prob > F	= 0.0000
				R-squared	= 0.2681
				Adj R-squared	= 0.2182
Total	99917052.6	188	531473.684	Root MSE	= 644.59

bwt	Coefficient	Std. err.	t	P>|t|	[95% conf. interval]
smoke	-293.3205	109.6431	-2.68	0.008	-509.7049 -76.93619
race					
2	-443.4656	149.8739	-2.96	0.004	-739.247 -147.6843
3	-300.2724	117.7204	-2.55	0.012	-532.576 -67.94725
age	-2.106441	9.651227	-0.22	0.827	-21.15347 16.94059

```
                                lwt |   4.458538   1.721434     2.59   0.010     1.061
> 229
>        7.855848
                                    |
                               ptl2 |
        One prior premature labor   |  -320.2626    149.579    -2.14   0.034    -615.4
> 618
>       -25.06333
Two or more prior premature labors  |   153.6952   277.8785     0.55   0.581    -394.7
> 076
>        702.0979
                                    |
                                 ht |  -575.0218    201.222    -2.86   0.005    -972.1
> 403
>       -177.9032
                                 ui |  -505.9775   137.8461    -3.67   0.000    -778.0
> 215
>       -233.9334
                                    |
                               ftv2 |
              One physician visit   |   118.7878   122.7752     0.97   0.335    -123.5
> 133
>        361.0888
           Two physician visits     |  -13.11499    139.511    -0.09   0.925    -288.4
> 447
>        262.2147
    Three or more physician visits  |  -112.8566    199.325    -0.57   0.572    -506.2
> 313
>        280.5181
                                    |
                              _cons |   2824.084   314.0448     8.99   0.000     2204.
> 305
>        3443.862
------------------------------------+----------------------------------------------
  ----------------
```

```
144
145
146 rvpplot smoke, yline(0)

147 *no pattern
148
149
150 regress bwt smoke race age lwt ptl2 ht ui ftv2
```

Source	SS	df	MS		
Model	22528099.7	8	2816012.47	Number of obs	= 189
Residual	77388952.9	180	429938.627	F(8, 180)	= 6.55
				Prob > F	= 0.0000
Total	99917052.6	188	531473.684	R-squared	= 0.2255
				Adj R-squared	= 0.1910
				Root MSE	= 655.7

bwt	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
smoke	-352.2265	107.4927	-3.28	0.001	-564.3345	-140.1186
race	-189.5989	57.89535	-3.27	0.001	-303.8398	-75.35806
age	.4082827	9.586487	0.04	0.966	-18.50807	19.32464
lwt	3.349241	1.687248	1.99	0.049	.0199116	6.67857
ptl2	-92.84892	108.0376	-0.86	0.391	-306.0321	120.3342
ht	-599.8101	203.596	-2.95	0.004	-1001.552	-198.0681
ui	-501.8176	139.5389	-3.60	0.000	-777.1602	-226.4751
ftv2	-26.56862	52.66522	-0.50	0.615	-130.4893	77.35201
_cons	3138.616	343.1829	9.15	0.000	2461.437	3815.795

```
151 rvpplot ptl2, yline(0)

152 * no pattern
153
154 regress bwt smoke i.race age lwt i.ptl2 ht ui i.ftv2
```

Source	SS	df	MS		
Model	26789683.1	12	2232473.59	Number of obs	189
Residual	73127369.5	176	415496.418	F(12, 176)	5.37
				Prob > F	0.0000
				R-squared	0.2681
				Adj R-squared	0.2182
Total	99917052.6	188	531473.684	Root MSE	644.59

bwt	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
smoke	-293.3205	109.6431	-2.68	0.008	-509.7049	-76.93619
race						
2	-443.4656	149.8739	-2.96	0.004	-739.2476	-147.6843
3	-300.2724	117.7204	-2.55	0.012	-532.5976	-67.94725
age	-2.106441	9.651227	-0.22	0.827	-21.15347	16.94059
lwt	4.458538	1.721434	2.59	0.010	1.061229	7.855848
ptl2						
One prior premature labor	-320.2626	149.579	-2.14	0.034	-615.4618	-25.06333
Two or more prior premature labors	153.6952	277.8785	0.55	0.581	-394.7076	702.0979
ht	-575.0218	201.222	-2.86	0.005	-972.1403	-177.9032
ui	-505.9775	137.8461	-3.67	0.000	-778.0215	-233.9334
ftv2						
One physician visit	118.7878	122.7752	0.97	0.335	-123.5133	361.0888
Two physician visits	-13.11499	139.511	-0.09	0.925	-288.4447	262.2147
Three or more physician visits	-112.8566	199.325	-0.57	0.572	-506.2313	280.5181
_cons	2824.084	314.0448	8.99	0.000	2204.305	3443.862

```
155
156
157 rvpplot ht, yline(0)

158 *no pattern
159
160 rvpplot ui, yline(0)

161 *no pattern
162
163 regress bwt smoke race age lwt ptl2 ht ui ftv2
```

Source	SS	df	MS		
Model	22528099.7	8	2816012.47	Number of obs	189
Residual	77388952.9	180	429938.627	F(8, 180)	6.55
				Prob > F	0.0000
Total	99917052.6	188	531473.684	R-squared	0.2255
				Adj R-squared	0.1910
				Root MSE	655.7

bwt	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
smoke	-352.2265	107.4927	-3.28	0.001	-564.3345	-140.1186
race	-189.5989	57.89535	-3.27	0.001	-303.8398	-75.35806
age	.4082827	9.586487	0.04	0.966	-18.50807	19.32464
lwt	3.349241	1.687248	1.99	0.049	.0199116	6.67857
ptl2	-92.84892	108.0376	-0.86	0.391	-306.0321	120.3342
ht	-599.8101	203.596	-2.95	0.004	-1001.552	-198.0681
ui	-501.8176	139.5389	-3.60	0.000	-777.1602	-226.4751
ftv2	-26.56862	52.66522	-0.50	0.615	-130.4893	77.35201
_cons	3138.616	343.1829	9.15	0.000	2461.437	3815.795

```
164 rvpplot ftv2, yline(0)

165 *ftv2
166
167 regress bwt smoke i.race age lwt i.ptl2 ht ui i.ftv2
```

Source	SS	df	MS		
Model	26789683.1	12	2232473.59	Number of obs	189
Residual	73127369.5	176	415496.418	F(12, 176)	5.37
				Prob > F	0.0000
Total	99917052.6	188	531473.684	R-squared	0.2681
				Adj R-squared	0.2182
				Root MSE	644.59

bwt	Coefficient	Std. err.	t	P>|t|	[95% conf. interval]
smoke	-293.3205	109.6431	-2.68	0.008	-509.7049 -76.93619
race					
2	-443.4656	149.8739	-2.96	0.004	-739.247 -147.6843
3	-300.2724	117.7204	-2.55	0.012	-532.5976 -67.94725
age	-2.106441	9.651227	-0.22	0.827	-21.15347 16.94059
lwt	4.458538	1.721434	2.59	0.010	1.061229 7.855848

```
                             ptl2 |
         One prior premature labor |  -320.2626   149.579    -2.14   0.034    -615.4
> 618
>       -25.06333
Two or more prior premature labors |   153.6952   277.8785     0.55   0.581    -394.7
> 076
>        702.0979
                                  |
                               ht |  -575.0218   201.222    -2.86   0.005    -972.1
> 403
>       -177.9032
                               ui |  -505.9775   137.8461    -3.67   0.000    -778.0
> 215
>       -233.9334
                                  |
                             ftv2 |
              One physician visit |   118.7878   122.7752     0.97   0.335    -123.5
> 133
>        361.0888
           Two physician visits |  -13.11499   139.511    -0.09   0.925    -288.4
> 447
>        262.2147
    Three or more physician visits |  -112.8566   199.325    -0.57   0.572    -506.2
> 313
>        280.5181
                                  |
                            _cons |   2824.084   314.0448     8.99   0.000     2204.
> 305
>        3443.862
------------------------------------------------------------------------------
```

```
168
169
170
171 * Interpretation: No clear pattern in any plot → independence assumption is met.
172
173
174 * CHECK NORMALITY OF RESIDUALS
175 * create new variable of residuals in the model
176 predict residuals, residuals

177
178
179 * Check normalty using:Shapiro-Wilk test.
180 swilk residuals
```

Shapiro-Wilk W test for normal data

Variable	Obs	W	V	z	Prob>z
residuals	189	0.99118	1.253	0.518	0.30222

```
181
182 * H0: Residuals are normally distributed.
183 * Non-significant p-value (p =  0.59724) means we fail to reject the null hypothesis
  >  that the residuals are normally distributed
184
185
186
```

```
187
188 * CHECK FOR EQUAL VARIANCE (HOMOSCEDASTICITY)
189
190 *Use both visual checks and statistical test
191
192 *Visual checks using rvpplot and rvf plots
193
194 * (a) RVP plots
195 rvpplot age, yline(0)

196 *distribution is rectangular
197
198 rvpplot lwt, yline(0)

199 *distribution is largely rectangular
200
201 regress bwt smoke race age lwt ptl2 ht ui ftv2
```

Source	SS	df	MS		
Model	22528099.7	8	2816012.47	Number of obs	= 189
Residual	77388952.9	180	429938.627	F(8, 180)	= 6.55
				Prob > F	= 0.0000
				R-squared	= 0.2255
				Adj R-squared	= 0.1910
Total	99917052.6	188	531473.684	Root MSE	= 655.7

bwt	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
smoke	-352.2265	107.4927	-3.28	0.001	-564.3345	-140.1186
race	-189.5989	57.89535	-3.27	0.001	-303.8398	-75.35806
age	.4082827	9.586487	0.04	0.966	-18.50807	19.32464
lwt	3.349241	1.687248	1.99	0.049	.0199116	6.67857
ptl2	-92.84892	108.0376	-0.86	0.391	-306.0321	120.3342
ht	-599.8101	203.596	-2.95	0.004	-1001.552	-198.0681
ui	-501.8176	139.5389	-3.60	0.000	-777.1602	-226.4751
ftv2	-26.56862	52.66522	-0.50	0.615	-130.4893	77.35201
_cons	3138.616	343.1829	9.15	0.000	2461.437	3815.795

```
202 rvpplot race, yline(0)

203 * no pattern
204
205 regress bwt smoke i.race age lwt i.ptl2 ht ui i.ftv2
```

Source	SS	df	MS		
Model	26789683.1	12	2232473.59	Number of obs	= 189
Residual	73127369.5	176	415496.418	F(12, 176)	= 5.37
				Prob > F	= 0.0000
				R-squared	= 0.2681
				Adj R-squared	= 0.2182
Total	99917052.6	188	531473.684	Root MSE	= 644.59

bwt	Coefficient	Std. err.	t	P>|t|	[95% conf. interval]
smoke	-293.3205	109.6431	-2.68	0.008	-509.7049 -76.93619
race					
2	-443.4656	149.8739	-2.96	0.004	-739.247 -147.6843
3	-300.2724	117.7204	-2.55	0.012	-532.5976 -67.94725
age	-2.106441	9.651227	-0.22	0.827	-21.15

```
> 347
>        16.94059
                                lwt |   4.458538   1.721434     2.59   0.010     1.061
> 229
>        7.855848
                                    |
                               ptl2 |
        One prior premature labor   |  -320.2626    149.579    -2.14   0.034    -615.4
> 618
>       -25.06333
Two or more prior premature labors  |   153.6952   277.8785     0.55   0.581    -394.7
> 076
>        702.0979
                                    |
                                 ht |  -575.0218    201.222    -2.86   0.005    -972.1
> 403
>       -177.9032
                                 ui |  -505.9775   137.8461    -3.67   0.000    -778.0
> 215
>       -233.9334
                                    |
                               ftv2 |
              One physician visit   |   118.7878   122.7752     0.97   0.335    -123.5
> 133
>        361.0888
           Two physician visits     |  -13.11499    139.511    -0.09   0.925    -288.4
> 447
>        262.2147
    Three or more physician visits  |  -112.8566    199.325    -0.57   0.572    -506.2
> 313
>        280.5181
                                    |
                              _cons |   2824.084   314.0448     8.99   0.000     2204.
> 305
>        3443.862
------------------------------------+----------------------------------------------------------------
  ----------------
```

```
206
207 *race
208
209 rvpplot smoke, yline(0)

210 *rectangular distribution is observed
211
212 regress bwt smoke race age lwt ptl2 ht ui ftv2
```

Source	SS	df	MS		
Model	22528099.7	8	2816012.47	Number of obs	= 189
Residual	77388952.9	180	429938.627	F(8, 180)	= 6.55
				Prob > F	= 0.0000
Total	99917052.6	188	531473.684	R-squared	= 0.2255
				Adj R-squared	= 0.1910
				Root MSE	= 655.7

bwt	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
smoke	-352.2265	107.4927	-3.28	0.001	-564.3345	-140.1186
race	-189.5989	57.89535	-3.27	0.001	-303.8398	-75.35806
age	.4082827	9.586487	0.04	0.966	-18.50807	19.32464
lwt	3.349241	1.687248	1.99	0.049	.0199116	6.67857
ptl2	-92.84892	108.0376	-0.86	0.391	-306.0321	120.3342
ht	-599.8101	203.596	-2.95	0.004	-1001.552	-198.0681
ui	-501.8176	139.5389	-3.60	0.000	-777.1602	-226.4751
ftv2	-26.56862	52.66522	-0.50	0.615	-130.4893	77.35201
_cons	3138.616	343.1829	9.15	0.000	2461.437	3815.795

```
213 rvpplot ptl2, yline(0)

214 * no pattern
215
216 regress bwt smoke i.race age lwt i.ptl2 ht ui i.ftv2
```

Source	SS	df	MS		
Model	26789683.1	12	2232473.59	Number of obs	189
Residual	73127369.5	176	415496.418	F(12, 176)	5.37
				Prob > F	0.0000
				R-squared	0.2681
				Adj R-squared	0.2182
Total	99917052.6	188	531473.684	Root MSE	644.59

bwt	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
smoke	-293.3205	109.6431	-2.68	0.008	-509.7049	-76.93619
race						
2	-443.4656	149.8739	-2.96	0.004	-739.2476	-147.6843
3	-300.2724	117.7204	-2.55	0.012	-532.5976	-67.94725
age	-2.106441	9.651227	-0.22	0.827	-21.15347	16.94059
lwt	4.458538	1.721434	2.59	0.010	1.061229	7.855848
ptl2						
One prior premature labor	-320.2626	149.579	-2.14	0.034	-615.4618	-25.06333
Two or more prior premature labors	153.6952	277.8785	0.55	0.581	-394.7076	702.0979
ht	-575.0218	201.222	-2.86	0.005	-972.1403	-177.9032
ui	-505.9775	137.8461	-3.67	0.000	-778.0215	-233.9334
ftv2						
One physician visit	118.7878	122.7752	0.97	0.335	-123.5133	361.0888
Two physician visits	-13.11499	139.511	-0.09	0.925	-288.4447	262.2147
Three or more physician visits	-112.8566	199.325	-0.57	0.572	-506.2313	280.5181
_cons	2824.084	314.0448	8.99	0.000	2204.305	3443.862

```
217
218
219 rvpplot ht, yline(0)

220 *rectangular distribution noted
221
222 rvpplot ui, yline(0)

223 *rectangular distribution noted
224
225
226 regress bwt smoke race age lwt ptl2 ht ui ftv2
```

Source	SS	df	MS		
Model	22528099.7	8	2816012.47	Number of obs	= 189
Residual	77388952.9	180	429938.627	F(8, 180)	= 6.55
				Prob > F	= 0.0000
Total	99917052.6	188	531473.684	R-squared	= 0.2255
				Adj R-squared	= 0.1910
				Root MSE	= 655.7

bwt	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
smoke	-352.2265	107.4927	-3.28	0.001	-564.3345	-140.1186
race	-189.5989	57.89535	-3.27	0.001	-303.8398	-75.35806
age	.4082827	9.586487	0.04	0.966	-18.50807	19.32464
lwt	3.349241	1.687248	1.99	0.049	.0199116	6.67857
ptl2	-92.84892	108.0376	-0.86	0.391	-306.0321	120.3342
ht	-599.8101	203.596	-2.95	0.004	-1001.552	-198.0681
ui	-501.8176	139.5389	-3.60	0.000	-777.1602	-226.4751
ftv2	-26.56862	52.66522	-0.50	0.615	-130.4893	77.35201
_cons	3138.616	343.1829	9.15	0.000	2461.437	3815.795

```
227 rvpplot ftv2, yline(0)

228 * no pattern
229
230 regress bwt smoke i.race age lwt i.ptl2 ht ui i.ftv2
```

Source	SS	df	MS		
Model	26789683.1	12	2232473.59	Number of obs	= 189
Residual	73127369.5	176	415496.418	F(12, 176)	= 5.37
				Prob > F	= 0.0000
Total	99917052.6	188	531473.684	R-squared	= 0.2681
				Adj R-squared	= 0.2182
				Root MSE	= 644.59

bwt	Coefficient	Std. err.	t	P>|t|	[95% conf. interval]
smoke	-293.3205	109.6431	-2.68	0.008	-509.7049 -76.93619
race					
2	-443.4656	149.8739	-2.96	0.004	-739.247 -147.6843
3	-300.2724	117.7204	-2.55	0.012	-532.5976 -67.94725
age	-2.106441	9.651227	-0.22	0.827	-21.15347 16.94059
lwt	4.458538	1.721434	2.59	0.010	1.061229 7.855848

```
                                 ptl2 |
         One prior premature labor | -320.2626   149.579    -2.14   0.034    -615.4
> 618
>       -25.06333
Two or more prior premature labors |  153.6952   277.8785     0.55   0.581    -394.7
> 076
>        702.0979
                                   |
                                  ht | -575.0218    201.222    -2.86   0.005    -972.1
> 403
>       -177.9032
                                  ui | -505.9775   137.8461    -3.67   0.000    -778.0
> 215
>       -233.9334
                                   |
                                ftv2 |
              One physician visit  |  118.7878   122.7752     0.97   0.335    -123.5
> 133
>        361.0888
             Two physician visits  | -13.11499    139.511    -0.09   0.925    -288.4
> 447
>        262.2147
    Three or more physician visits | -112.8566    199.325    -0.57   0.572    -506.2
> 313
>        280.5181
                                   |
                              _cons |  2824.084   314.0448     8.99   0.000     2204.
> 305
>        3443.862
-----------------------------------+----------------------------------------------------------------
  ----------------
```

```
231
232 * (b) RVF plot
233 * First predict fitted values
234 predict fitted, xb

235
236 rvfplot, yline(0)

237 *by observation; we can see that there is a a rectangular spread of the data points
  > ; this fulfills the assumption of equal variance
238
239 * Use Statistical test: Breusch-Pagan / Cook-Weisberg test (estat hettest)
240
241 * H0: Constant variance. p > 0.05 → assumption holds.
242
243 * Overall check
244 estat hettest

  Breusch-Pagan/Cook-Weisberg test for heteroskedasticity
  Assumption: Normal error terms
  Variable: Fitted values of bwt

  H0: Constant variance

      chi2(1) =   0.03
  Prob > chi2 = 0.8660
```

```
245
246 * per variable;
247
248 estat hettest age

  Breusch-Pagan/Cook-Weisberg test for heteroskedasticity
  Assumption: Normal error terms
  Variable: age

  H0: Constant variance

      chi2(1) =   5.21
  Prob > chi2 = 0.0224

249 *Breusch-Pagan / Cook-Weisberg test for age is signficant; hence null hypotheis of c
  > onstant variance is rejected
250
251 estat hettest race

  Breusch-Pagan/Cook-Weisberg test for heteroskedasticity
  Assumption: Normal error terms
  Variable: race

  H0: Constant variance

      chi2(1) =   0.02
  Prob > chi2 = 0.8852

252 estat hettest smoke

  Breusch-Pagan/Cook-Weisberg test for heteroskedasticity
  Assumption: Normal error terms
  Variable: smoke

  H0: Constant variance

      chi2(1) =   0.53
  Prob > chi2 = 0.4682

253 estat hettest lwt

  Breusch-Pagan/Cook-Weisberg test for heteroskedasticity
  Assumption: Normal error terms
  Variable: lwt

  H0: Constant variance

      chi2(1) =   0.16
  Prob > chi2 = 0.6923

254 estat hettest ptl2

  Breusch-Pagan/Cook-Weisberg test for heteroskedasticity
  Assumption: Normal error terms
  Variable: ptl2

  H0: Constant variance

      chi2(1) =   0.12
  Prob > chi2 = 0.7294
```

```
255 estat hettest ht

  Breusch-Pagan/Cook-Weisberg test for heteroskedasticity
  Assumption: Normal error terms
  Variable: ht

  H0: Constant variance

      chi2(1) =   0.55
  Prob > chi2 = 0.4563

256 estat hettest ui

  Breusch-Pagan/Cook-Weisberg test for heteroskedasticity
  Assumption: Normal error terms
  Variable: ui

  H0: Constant variance

      chi2(1) =   0.95
  Prob > chi2 = 0.3300

257 estat hettest ftv2

  Breusch-Pagan/Cook-Weisberg test for heteroskedasticity
  Assumption: Normal error terms
  Variable: ftv2

  H0: Constant variance

      chi2(1) =   0.06
  Prob > chi2 = 0.8077

258
259 *Though the age-specific Breusch-Pagan / Cook-Weisberg test for age is signficant, t
  > he overall test is non signficant and the RVP plot for age looked rectangular; there
  > fore the  assumption of equal variance in model is considered met.
260
261
262 *CHECK FOR MULTICOLLINEARITY
263 * Correlation matrix - eyeball for strong relationships
264
265 correlate smoke race lwt ptl2 ht ui age ftv2 bwt
  (obs=189)
             |    smoke     race      lwt     ptl2       ht       ui      age     ftv2
  >       bwt
-------------+------------------------------------------------------------------------
  ___________
       smoke |   1.0000
        race |  -0.3390   1.0000
         lwt |  -0.0442  -0.1650   1.0000
        ptl2 |   0.1837   0.0189  -0.1347   1.0000
          ht |   0.0134   0.0199   0.2364  -0.0133   1.0000
          ui |   0.0622   0.0536  -0.1528   0.2128  -0.1086   1.0000
         age |  -0.0443  -0.1728   0.1801   0.0717  -0.0158  -0.0752   1.0000
        ftv2 |  -0.0460  -0.1295   0.0884  -0.0512  -0.0711  -0.0505   0.2184   1.0000
         bwt |  -0.1891  -0.1962   0.1858  -0.1740  -0.1461  -0.2835   0.0899   0.0503
  >    1.0000
```

```
266 *from  eyeballing; there was no multicollinearity as the relationship between contin
  > uous predictors was not strong (0.1801)
267
268
269
270 * Using VIF (Variance Inflation Factor)
271
272 estat vif
```

Variable	VIF	1/VIF
smoke	**1.30**	**0.767604**
race		
2	**1.21**	**0.824928**
3	**1.44**	**0.693252**
age	**1.18**	**0.845102**
lwt	**1.25**	**0.797575**
ptl2		
1	**1.13**	**0.886325**
2	**1.08**	**0.926226**
ht	**1.10**	**0.913111**
ui	**1.09**	**0.916761**
ftv2		
1	**1.28**	**0.780588**
2	**1.18**	**0.845851**
3	**1.07**	**0.930575**
Mean VIF	**1.19**	

```
273 *No VIF was noted >10; No multicollinearity between the predictors
274
275
276 * CHECK FOR OUTLIERS
277
278 * Using DFbetas
279
280 predict dfbetas_age, dfbeta(age)

281 predict dfbeta_smoke, dfbeta(smoke)

282 predict dfbeta_2race, dfbeta(2.race)

283 predict dfbeta_3race, dfbeta(3.race)

284 predict dfbeta_lwt, dfbeta(lwt)

285 predict dfbeta_1ptl2, dfbeta(1.ptl2)

286 predict dfbeta_2ptl2, dfbeta(2.ptl2)

287 predict dfbeta_ht, dfbeta(ht)

288 predict dfbeta_ui, dfbeta(ui)

289 predict dfbeta_1ftv2, dfbeta(1.ftv2)

290 predict dfbeta_2ftv2, dfbeta(2.ftv2)

291 predict dfbeta_3ftv2, dfbeta(3.ftv2)
```

```
292
293
294 * Summarise each ;
295
296 sum dfbetas_age
```

Variable	Obs	Mean	Std. dev.	Min	Max
dfbetas_age	**189**	**.0001922**	**.0997504**	**-.4358608**	**.8462367**

```
297 * noted to have max dfbeta > 0.5
298
299 *to identify ids causing this influence ;
300
301 list if abs( dfbetas_age) > 0.5
```

	id	age	lwt	race	smoke	ptl	ht	ui	ftv	bwt	ptl2	ftv2	residu~s	fitted	dfbet~ge	dfbeta~ke	df~2race	dfb~3race	dfbeta~wt	dfb~1ptl2	dfb~2ptl2	dfbet~ht	dfbeta~i	df~1ftv2	dfb~2ftv2	dfb~3ftv2
130.	**226**	**45**	**123**	**1**	**0**	**0**	**0**	**0**	**1**	**4990**	**No prior prematur e labor**	**One physician visit**	**1593.518**	**3396.482**	**.8462367**	**-.0946698**	**.057853 2**	**-.0919594**	**-.2277828**	**-.2031392**	**-.0771283**	**.0001991**	**.0065741**	**.1235413**	**-.1709331**	**-.0947763**

```
302 * ids listed are 159,188 and 226
303
304
305 sum dfbeta_smoke
```

Variable	Obs	Mean	Std. dev.	Min	Max
dfbeta_smoke	**189**	**-.0000483**	**.0749203**	**-.243478**	**.2588947**

```
306 *both abs min max dfbeta are <0.5
307
308 sum dfbeta_2race
```

Variable	Obs	Mean	Std. dev.	Min	Max
dfbeta_2race	**189**	**-.0002255**	**.0694604**	**-.4112781**	**.2672275**

```
309 *both abs min max dfbeta are <0.5
310
311 sum dfbeta_3race
```

Variable	Obs	Mean	Std. dev.	Min	Max
dfbeta_3race	**189**	**-.000305**	**.0770598**	**-.2204692**	**.2534211**

```
312 *both abs min max dfbeta are <0.5
313
314
315 sum dfbeta_lwt
```

Variable	Obs	Mean	Std. dev.	Min	Max
dfbeta_lwt	**189**	**.0003746**	**.0731693**	**-.2465397**	**.2912798**

```
316 *both abs min max dfbeta are <0.5
317
318
319 sum dfbeta_1ptl2
```

Variable	Obs	Mean	Std. dev.	Min	Max
dfbeta_1ptl2	**189**	**.0002136**	**.0793465**	**-.3182063**	**.3904274**

```
320 *both abs min max dfbeta are <0.5
321
322
323 sum dfbeta_2ptl2
```

Variable	Obs	Mean	Std. dev.	Min	Max
dfbeta_2ptl2	**189**	**-.0000742**	**.0970423**	**-.8761988**	**.7555826**

```
324 *both abs min max dfbeta are <0.5
325
326
327 sum dfbeta_ht
```

Variable	Obs	Mean	Std. dev.	Min	Max
dfbeta_ht	**189**	**-.0001866**	**.0864127**	**-.5336523**	**.4472942**

```
328 *min abs dfbeta>0.5
329
330 **to identify ids causing this influence ;
331
332 list if abs(dfbeta_ht) > 0.5
```

	id	age	lwt	race	smoke	ptl	ht	ui	ftv	bwt	ptl2	ftv2	residuals	fitted	dfbetas~e	dfbet~ke	dfb~2race	dfb~3race	dfbeta~wt	df~1ptl2	df~2ptl2	df~3ftv2	dfbeta_ht	dfbeta_ui	df~1ftv2	df~2ftv2
133.	**11**	**34**	**187**	**2**	**1**	**0**	**1**	**0**	**0**	**1135**	**No prior premature labor**	**No physician visits**	**-1139.403**	**2274.403**	**-.4358608**	**-.243478**	**-.41127**	**-.1343776**	**-.0235547**	**.1785985**	**.0155561**	**-.**	**-.5336523**	**-.0347399**	**.1016487**	**.0921492**
																						.1283023				

```
333 ** ids listed are 159,188 and 11
334
335
336 sum dfbeta_ui
```

Variable	Obs	Mean	Std. dev.	Min	Max
dfbeta_ui	**189**	**.0001852**	**.0854952**	**-.5757909**	**.3582423**

```
337 *min abs dfbeta>0.5
338
339 **to identify ids causing this influence ;
340
341 list if abs(dfbeta_ui) > 0.5
```

	id	age	lwt	race	smoke	ptl	ht	ui	ftv	bwt	ptl2	ftv2	residuals	fitted	dfbetas~e	dfbet~ke	df~2race
132.	**10**	**29**	**130**	**1**	**0**	**0**	**0**	**1**	**2**	**1021**	**No prior premature labor**	**Two physician visits**	**-1802.514**	**2823.514**	**-.1977893**	**.2588947**	**.1146002**

	df~3race	dfbet~wt	df~1ptl2	df~2ptl2	dfbeta_ht	dfbeta_ui	df~1ftv2	dfb~2ftv2	dfb~3ftv2
	.2534211	**.0699314**	**.0793891**	**.0969735**	**-.021431**	**-.5757909**	**.0807833**	**-.3960064**	**-.0059288**

```
342 *ids are 159,188,10
343
344 sum dfbeta_1ftv2
```

Variable	Obs	Mean	Std. dev.	Min	Max
dfbeta_1ftv2	**189**	**-.0002647**	**.0785257**	**-.3359247**	**.237453**

```
345 *both abs min max dfbeta are <0.5
346
347
348 sum dfbeta_2ftv2
```

Variable	Obs	Mean	Std. dev.	Min	Max
dfbeta_2ftv2	**189**	**-.0003227**	**.0778068**	**-.3960064**	**.2400846**

```
349 *both abs min max dfbeta are <0.5
350
351
352 sum dfbeta_3ftv2
```

Variable	Obs	Mean	Std. dev.	Min	Max
dfbeta_3ftv2	**189**	**.0001181**	**.0605618**	**-.245884**	**.437811**

```
353 *both abs min max dfbeta are <0.5
354
355
356
357 regress bwt smoke i.race age lwt i.ptl2 ht ui i.ftv2
```

Source	SS	df	MS		
Model	26789683.1	12	2232473.59	Number of obs	= 189
Residual	73127369.5	176	415496.418	F(12, 176)	= 5.37
				Prob > F	= 0.0000
				R-squared	= 0.2681
				Adj R-squared	= 0.2182
Total	99917052.6	188	531473.684	Root MSE	= 644.59

bwt	Coefficient	Std. err.	t	P>|t|	[95% conf. interval]
smoke	-293.3205	109.6431	-2.68	0.008	-509.7049 -76.93619
race					
2	-443.4656	149.8739	-2.96	0.004	-739.247 -147.6843
3	-300.2724	117.7204	-2.55	0.012	-532.5976 -67.94725
age	-2.106441	9.651227	-0.22	0.827	-21.15347 16.94059
lwt	4.458538	1.721434	2.59	0.010	1.061229 7.855848
ptl2					
One prior premature labor	-320.2626	149.579	-2.14	0.034	-615.4618 -25.06333
Two or more prior premature labors	153.6952	277.8785	0.55	0.581	-394.7076 702.0979
ht	-575.0218	201.222	-2.86	0.005	-972.1403 -177.9032
ui	-505.9775	137.8461	-3.67	0.000	-778.0215 -233.9334
ftv2					
One physician visit	118.7878	122.7752	0.97	0.335	-123.5133 361.0888
Two physician visits	-13.11499	139.511	-0.09	0.925	-288.4447 262.2147
Three or more physician visits	-112.8566	199.325	-0.57	0.572	-506.2313 280.5181
_cons	2824.084	314.0448	8.99	0.000	2204.305 3443.862

```
358
359
360 * Using Cook's Distance
361
362 predict cooksd, cooksd

363 summarize cooksd
```

Variable	Obs	Mean	Std. dev.	Min	Max
cooksd	**189**	**.00642**	**.0117865**	**9.75e-07**	**.0738859**

```
364 *max is <1
365
366
367
368
369 *What predictors enter the multivariate analysis
370 *we shall do simple linear regression for each predictor outcome and set cut off of
  > p-value<0.2
371
372 regress bwt age
```

Source	SS	df	MS		
				Number of obs	= **189**
				F(1, 187)	= **1.52**
Model	**806926.913**	**1**	**806926.913**	Prob > F	= **0.2188**
Residual	**99110125.7**	**187**	**530000.672**	R-squared	= **0.0081**
				Adj R-squared	= **0.0028**
Total	**99917052.6**	**188**	**531473.684**	Root MSE	= **728.01**

bwt	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
age	**12.36433**	**10.02055**	**1.23**	**0.219**	**-7.403527**	**32.13219**
_cons	**2657.333**	**238.804**	**11.13**	**0.000**	**2186.236**	**3128.429**

```
373 *pvalue= 0.219
374
375 regress bwt lwt
```

Source	SS	df	MS		
				Number of obs	= **189**
				F(1, 187)	= **6.69**
Model	**3448881.3**	**1**	**3448881.3**	Prob > F	= **0.0105**
Residual	**96468171.3**	**187**	**515872.574**	R-squared	= **0.0345**
				Adj R-squared	= **0.0294**
Total	**99917052.6**	**188**	**531473.684**	Root MSE	= **718.24**

bwt	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
lwt	**4.429264**	**1.713025**	**2.59**	**0.010**	**1.049927**	**7.8086**
_cons	**2369.672**	**228.4306**	**10.37**	**0.000**	**1919.04**	**2820.304**

```
376 *pvalue=0.010
377
378 regress bwt i.race
```

Source	SS	df	MS		
				Number of obs	= **189**
				F(2, 186)	= **4.97**
Model	**5070607.63**	**2**	**2535303.82**	Prob > F	= **0.0079**
Residual	**94846445**	**186**	**509927.124**	R-squared	= **0.0507**
				Adj R-squared	= **0.0405**
Total	**99917052.6**	**188**	**531473.684**	Root MSE	= **714.09**

bwt	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
race						
2	-384.0473	157.8744	-2.43	0.016	-695.5019	-72.59266
3	-299.7247	113.6776	-2.64	0.009	-523.9878	-75.4615
_cons	3103.74	72.88169	42.59	0.000	2959.959	3247.521

```
379 *pvalue=0.0079
380
381 regress bwt smoke
```

Source	SS	df	MS		
Model	3573406.24	1	3573406.24	Number of obs	= 189
Residual	96343646.4	187	515206.665	F(1, 187)	= 6.94
				Prob > F	= 0.0092
Total	99917052.6	188	531473.684	R-squared	= 0.0358
				Adj R-squared	= 0.0306
				Root MSE	= 717.78

bwt	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
smoke	-281.7133	106.9687	-2.63	0.009	-492.7338	-70.69274
_cons	3054.957	66.93324	45.64	0.000	2922.915	3186.998

```
382 *pvalue= 0.009
383
384 regress bwt i.ptl2
```

Source	SS	df	MS		
Model	5586361.94	2	2793180.97	Number of obs	= 189
Residual	94330690.7	186	507154.251	F(2, 186)	= 5.51
				Prob > F	= 0.0047
Total	99917052.6	188	531473.684	R-squared	= 0.0559
				Adj R-squared	= 0.0458
				Root MSE	= 712.15

bwt	Coefficient	Std. err.	t	P>|t|	[95% conf. interval]
ptl2					
One prior premature labor	-517.2807	155.9522	-3.32	0.001	-824.9431 -209.6182
Two or more prior premature labors	-101.739	296.1678	-0.34	0.732	-686.0188 482.5408
_cons	3013.572	56.47698	53.36	0.000	2902.155 3124.99

```
385 *pvalue=0.0047
386
387 regress bwt ht
```

Source	SS	df	MS		
Model	2132014.49	1	2132014.49	Number of obs	= 189
Residual	97785038.2	187	522914.643	F(1, 187)	= 4.08
				Prob > F	= 0.0449
				R-squared	= 0.0213
				Adj R-squared	= 0.0161
Total	99917052.6	188	531473.684	Root MSE	= 723.13

bwt	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
ht	-435.5607	215.7094	-2.02	0.045	-861.0973	-10.02413
_cons	2972.311	54.35366	54.68	0.000	2865.086	3079.536

```
388 *pvalue= 0.045
389
390 regress bwt ui
```

Source	SS	df	MS		
Model	8028747.44	1	8028747.44	Number of obs	= 189
Residual	91888305.2	187	491381.311	F(1, 187)	= 16.34
				Prob > F	= 0.0001
				R-squared	= 0.0804
				Adj R-squared	= 0.0754
Total	99917052.6	188	531473.684	Root MSE	= 700.99

bwt	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
ui	-580.1801	143.5318	-4.04	0.000	-863.3298	-297.0304
_cons	3030.609	55.24543	54.86	0.000	2921.624	3139.593

```
391 *pvalue= 0.000
392
393 regress bwt i.ftv2
```

Source	SS	df	MS		
Model	2256992.02	3	752330.673	Number of obs	= 189
Residual	97660060.6	185	527892.22	F(3, 185)	= 1.43
				Prob > F	= 0.2369
				R-squared	= 0.0226
				Adj R-squared	= 0.0067
Total	99917052.6	188	531473.684	Root MSE	= 726.56

bwt	Coefficient	Std. err.	t	P>|t|	[95% conf. interval]
ftv2					
One physician visit	242.73	128.4939	1.89	0.060	-10.77167 496.2317
Two physician visits	145.0633	151.2459	0.96	0.339	-153.3252 443.4519
Three or more physician visits	-63.02	221.9683	-0.28	0.777	-500.9347 374.8947
_cons	2865.27	72.65619	39.44	0.000	2721.929 3008.611

```
394 *pvalue= 0.2369
395
396 *because Age of the Mother in Years and Number of Physician Visits During the First
  > Trimester have a p value >0.2; they should be dropped from the model; However ; mate
  > rnal age is clinically important so it cannot be dropped
397
398
399 *
400 * STEP 10: CHECK FOR INTERACTION
401 * Main predictors: smoke and race.
402
403 **check for signficance of variables
404 regress bwt age lwt i.race smoke i.ptl2 ht ui
```

Source	SS	df	MS		
Model	**26110508**	**9**	**2901167.56**	Number of obs	= **189**
Residual	**73806544.6**	**179**	**412327.065**	F(9, 179)	= **7.04**
				Prob > F	= **0.0000**
				R-squared	= **0.2613**
				Adj R-squared	= **0.2242**
Total	**99917052.6**	**188**	**531473.684**	Root MSE	= **642.13**

bwt	Coefficient	Std. err.	t	P>|t|	[95% conf. interval]
age	**-1.4796**	**9.409666**	**-0.16**	**0.875**	**-20.04774** **17.08854**
lwt	**4.230848**	**1.703954**	**2.48**	**0.014**	**.8684271** **7.593269**
race					
2	**-453.6814**	**149.0072**	**-3.04**	**0.003**	**-747.7181** **-159.6447**
3	**-327.286**	**113.8901**	**-2.87**	**0.005**	**-552.3026** **-102.5459**
smoke	**-325.4022**	**105.4711**	**-3.09**	**0.002**	**-533.5289** **-117.2754**
ptl2					
One prior premature labor	**-295.8071**	**146.966**	**-2.01**	**0.046**	**-585.8159** **-5.798431**
Two or more prior premature labors	**183.1224**	**275.106**	**0.67**	**0.506**	**-359.7458** **725.9907**
ht	**-567.9902**	**198.9717**	**-2.85**	**0.005**	**-960.6221** **-175.3583**
ui	**-514.5269**	**136.7178**	**-3.76**	**0.000**	**-784.3129** **-244.741**
_cons	**2879.69**	**309.5318**	**9.30**	**0.000**	**2268.889** **3490.491**

```
405 *age is not signficant; so we dont from interaction terms with its
406
407 *
408
409 * Use stepwise backward to remove non-significant interaction terms.
410 *
411 stepwise, pr(0.05) : regress bwt (i.smoke i.race) i.smoke#i.race i.smoke##c.lwt i.sm
  > oke##ht i.smoke##ui i.smoke##i.ptl2 i.race##c.lwt i.race##i.ht i.race##ui i.race##i.
  > ptl2
  note: 0b.smoke omitted because of estimability.
  note: 1b.race omitted because of estimability.
  note: 0o.smoke#1o.race omitted because of estimability.
  note: 0.smoke#2.race omitted because of estimability.
  note: 0.smoke#3.race omitted because of estimability.
  note: 1.smoke#1.race omitted because of estimability.
  note: lwt omitted because of estimability.
  note: 0o.smoke#co.lwt omitted because of estimability.
  note: 0b.ht omitted because of estimability.
  note: 0o.smoke#0o.ht omitted because of estimability.
  note: 0o.smoke#1o.ht omitted because of estimability.
  note: 1o.smoke#0o.ht omitted because of estimability.
  note: 0b.ui omitted because of estimability.
  note: 0o.smoke#0o.ui omitted because of estimability.
  note: 0o.smoke#1o.ui omitted because of estimability.
  note: 1o.smoke#0o.ui omitted because of estimability.
  note: 0b.ptl2 omitted because of estimability.
  note: 0o.smoke#0o.ptl2 omitted because of estimability.
  note: 0o.smoke#1o.ptl2 omitted because of estimability.
  note: 0o.smoke#2o.ptl2 omitted because of estimability.
  note: 1o.smoke#0o.ptl2 omitted because of estimability.
  note: 1o.race#co.lwt omitted because of estimability.
  note: 1o.race#0o.ht omitted because of estimability.
  note: 1o.race#1o.ht omitted because of estimability.
  note: 2o.race#0o.ht omitted because of estimability.
  note: 3o.race#0o.ht omitted because of estimability.
  note: 1o.race#0o.ui omitted because of estimability.
  note: 1o.race#1o.ui omitted because of estimability.
  note: 2o.race#0o.ui omitted because of estimability.
  note: 3o.race#0o.ui omitted because of estimability.
  note: 1o.race#0o.ptl2 omitted because of estimability.
  note: 1o.race#1o.ptl2 omitted because of estimability.
  note: 1o.race#2o.ptl2 omitted because of estimability.
  note: 2o.race#0o.ptl2 omitted because of estimability.
  note: 2o.race#2o.ptl2 omitted because of estimability.
  note: 3o.race#0o.ptl2 omitted because of estimability.

  Wald test, begin with full model:
  p = 0.9779 >= 0.0500, removing 1.smoke#c.lwt
  p = 0.8760 >= 0.0500, removing 3.race#1.ui
  p = 0.8058 >= 0.0500, removing 2.ptl2
  p = 0.8058 >= 0.0500, removing 2.ptl2
  p = 0.6302 >= 0.0500, removing 2.race#1.ptl2
  p = 0.6317 >= 0.0500, removing 1.smoke#2.race
  p = 0.6456 >= 0.0500, removing 1.ptl2
  p = 0.6456 >= 0.0500, removing 1.ptl2
  p = 0.6205 >= 0.0500, removing 1.smoke#1.ui
  p = 0.5660 >= 0.0500, removing 1.ht
  p = 0.5660 >= 0.0500, removing 1.ht
  p = 0.4907 >= 0.0500, removing 3.race#1.ptl2
  p = 0.3392 >= 0.0500, removing 1.smoke#1.ht
  p = 0.3071 >= 0.0500, removing 3.race#2.ptl2
  p = 0.3974 >= 0.0500, removing 1.smoke#2.ptl2
  p = 0.1919 >= 0.0500, removing 2.race#1.ui
  p = 0.2205 >= 0.0500, removing 2.race#c.lwt
  p = 0.2856 >= 0.0500, removing 2.race#1.ht
  p = 0.0911 >= 0.0500, removing 3.race#c.lwt
```

Source	SS	df	MS	Number of obs	=	189
				F(7, 181)	=	9.50
Model	26847686.2	7	3835383.75	Prob > F	=	0.0000
Residual	73069366.4	181	403698.157	R-squared	=	0.2687
				Adj R-squared	=	0.2404
Total	99917052.6	188	531473.684	Root MSE	=	635.37

bwt	Coefficient	Std. err.	t	P>|t|	[95% conf. interval]
1.smoke	-449.9509	120.8418	-3.72	0.000	-688.3907 -211.511
race					
2	-473.6543	141.6819	-3.34	0.001	-753.215 -194.0936
3	-506.6406	126.859	-3.99	0.000	-756.9532 -256.3279
smoke#race					
1 3	548.7293	235.6858	2.33	0.021	83.68424 1013.774
1.ui	-549.9903	132.1732	-4.16	0.000	-810.7888 -289.1918
race#ht					
3 1	-896.6995	330.5602	-2.71	0.007	-1548.947 -244.4523
smoke#ptl2					
1#One prior premature labor	-470.9246	189.7619	-2.48	0.014	-845.3546 -96.49456
_cons	3466.09	91.71842	37.79	0.000	3285.115 3647.065

```
412
413 *then we bring back the predictors
414
415 regress bwt i.smoke i.race i.smoke#i.race i.race#i.ht i.smoke#i.ptl2 i.ht i.ui i.ptl
  > 2
```

Source	SS	df	MS	Number of obs	=	189
				F(13, 175)	=	5.22
Model	27935093.8	13	2148853.37	Prob > F	=	0.0000
Residual	71981958.8	175	411325.479	R-squared	=	0.2796
				Adj R-squared	=	0.2261
Total	99917052.6	188	531473.684	Root MSE	=	641.35

bwt	Coefficient	Std. err.	t	P>|t|	[95% conf. interval]
1.smoke	-488.3979	139.7189	-3.50	0.001	-764.1488 -212.647
race					
2	-476.8397	196.1984	-2.43	0.016	-864.0593 -89.6201
3	-522.2059	133.4373	-3.91	0.000	-785.5593

```
>         -258.8524
                                        |
                          smoke#race    |
                                1 2     |   114.9077   294.638     0.39   0.697    -466
> .5935
>          696.4089
                                1 3     |   557.8333   245.9875    2.27   0.025      72
> .3492
>          1043.317
                                        |
                            race#ht     |
                                2 1     |  -270.0486   494.2018   -0.55   0.585    -124
> 5.411
>          705.3143
                                3 1     |  -756.8714   456.7031   -1.66   0.099    -165
> 8.226
>          144.4836
                                        |
                          smoke#ptl2    |
        1#One prior premature labor     |  -461.0504   295.5487   -1.56   0.121    -104
> 4.349
>          122.2482
1#Two or more prior premature labors    |   557.1381   567.5967    0.98   0.328    -563
> .0778
>          1677.354
                                        |
                                1.ht    |  -144.4255   299.3605   -0.48   0.630    -735
> .2471
>          446.3961
                                1.ui    |  -561.4512   137.2283   -4.09   0.000    -832
> .2867
>         -290.6157
                                        |
                                ptl2    |
          One prior premature labor     |  -5.782161   224.2486   -0.03   0.979     -44
> 8.362
>          436.7977
  Two or more prior premature labors    |  -261.3153   461.6875   -0.57   0.572    -117
> 2.507
>          649.8769
                                        |
                               _cons    |   3489.144   98.30021   35.49   0.000     329
> 5.137
>           3683.15
------------------------------------------------------------------------------
  ------------------
```

```
416 *notice
417
418 *final model after interaction is as follows;
419
420 regress bwt i.smoke i.race i.smoke#i.race i.ui
```

Source	SS	df	MS		
Model	21410334.2	6	3568389.03	Number of obs	= 189
Residual	78506718.4	182	431355.596	F(6, 182)	= 8.27
				Prob > F	= 0.0000
				R-squared	= 0.2143
				Adj R-squared	= 0.1884
Total	99917052.6	188	531473.684	Root MSE	= 656.78

bwt	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
1.smoke	-554.8017	135.0013	-4.11	0.000	-821.1706	-288.4327
race						
2	-521.0953	192.1929	-2.71	0.007	-900.308	-141.8827
3	-584.4969	133.0493	-4.39	0.000	-847.0143	-321.9794
smoke#race						
1 2	101.1191	299.2605	0.34	0.736	-489.347	691.5851
1 3	601.123	249.1934	2.41	0.017	109.4434	1092.803

1.ui	-550.3073	136.8903	-4.02	0.000	-820.4033	-280.2113
_cons	3478.778	99.7918	34.86	0.000	3281.88	3675.676

```
421
422
423 *CHECK FOR CONFOUNDING;
424 *we start with last variable to be dropped ;
425
426 *hypertension (ht)
427
428 regress bwt i.smoke i.race i.smoke#i.race i.ui ht
```

Source	SS	df	MS		
Model	23463582.1	7	3351940.29	Number of obs	= 189
Residual	76453470.6	181	422394.865	F(7, 181)	= 7.94
				Prob > F	= 0.0000
Total	99917052.6	188	531473.684	R-squared	= 0.2348
				Adj R-squared	= 0.2052
				Root MSE	= 649.92

bwt	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
1.smoke	-528.5806	134.1201	-3.94	0.000	-793.2205	-263.9406
race						
2	-473.532	191.4057	-2.47	0.014	-851.2056	-95.85841
3	-561.0034	132.0906	-4.25	0.000	-821.6389	-300.3679
smoke#race						
1 2	57.89384	296.7841	0.20	0.846	-527.7078	643.4955
1 3	549.4975	247.7008	2.22	0.028	60.74499	1038.25
1.ui	-583.1192	136.276	-4.28	0.000	-852.0133	-314.2252
ht	-434.0746	196.8806	-2.20	0.029	-822.5511	-45.59819
_cons	3491.626	98.92165	35.30	0.000	3296.438	3686.814

```
429
430 *History of Premature Labor (ptl2)
431
432 regress bwt i.smoke i.race i.smoke#i.race i.ui i.ptl2
```

Source	SS	df	MS		
Model	23667931.3	8	2958491.42	Number of obs	= 189
Residual	76249121.3	180	423606.23	F(8, 180)	= 6.98
				Prob > F	= 0.0000
Total	99917052.6	188	531473.684	R-squared	= 0.2369
				Adj R-squared	= 0.2030
				Root MSE	= 650.85

bwt	Coefficient	Std. err.	t	P>|t|	[95% conf. interval]
1.smoke	-511.9453	135.7706	-3.77	0.000	-779.852 -244.0386
race					
2	-486.8832	191.0838	-2.55	0.012	-863.957 -109.8307
3	-551.078	132.6534	-4.15	0.000	-812.839 -289.3222
smoke#race					
1 2	85.73506	296.6894	0.29	0.773	-499.7

```
> 015
>        671.1716
                              1 3 |   586.6353   247.0615     2.37   0.019     99.12
> 598
>        1074.145
                                  |
                              1.ui |  -533.0904   138.3795    -3.85   0.000    -806.1
> 451
>       -260.0357
                                  |
                             ptl2 |
         One prior premature labor |  -323.2567   146.4491    -2.21   0.029    -612.2
> 345
>       -34.27883
Two or more prior premature labors |   123.8508   277.8927     0.45   0.656    -424.4
> 957
>        672.1973
                                  |
                            _cons |   3481.745   99.01526    35.16   0.000     3286.
> 365
>        3677.125
------------------------------------+----------------------------------------------------------------
```

```
433
434 *Weight in Pounds at the Last Menstrual Period(lwt)
435
436 regress bwt i.smoke i.race i.smoke#i.race i.ui lwt
```

Source	SS	df	MS		
Model	22518035.4	7	3216862.2	Number of obs	= 189
Residual	77399017.2	181	427618.88	F(7, 181)	= 7.52
				Prob > F	= 0.0000
				R-squared	= 0.2254
				Adj R-squared	= 0.1954
Total	99917052.6	188	531473.684	Root MSE	= 653.93

bwt	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
1.smoke	-523.7502	135.7928	-3.86	0.000	-791.6908	-255.8096
race						
2	-552.5264	192.3525	-2.87	0.005	-932.068	-172.9847
3	-533.3755	136.2265	-3.92	0.000	-802.1717	-264.5793
smoke#race						
1 2	94.2514	297.992	0.32	0.752	-493.7336	682.2364
1 3	551.1941	250.0436	2.20	0.029	57.81884	1044.569
1.ui	-519.1285	137.6659	-3.77	0.000	-790.7649	-247.492
lwt	2.681943	1.666351	1.61	0.109	-.6060304	5.969915
_cons	3103.58	253.4098	12.25	0.000	2603.563	3603.598

```
437
438 *Age
439
440 regress bwt i.smoke i.race i.smoke#i.race i.ui age
```

Source	SS	df	MS		
Model	21447887	7	3063983.86	Number of obs	= 189
Residual	78469165.7	181	433531.302	F(7, 181)	= 7.07
				Prob > F	= 0.0000
				R-squared	= 0.2147
				Adj R-squared	= 0.1843
Total	99917052.6	188	531473.684	Root MSE	= 658.43

bwt	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
1.smoke	-563.7015	138.6784	-4.06	0.000	-837.3358	-290.0673
race						
2	-538.1467	201.1989	-2.67	0.008	-935.1438	-141.1497
3	-594.7562	137.8641	-4.31	0.000	-866.7838	-322.7286
smoke#race						
1 2	121.4699	307.8795	0.39	0.694	-486.0248	728.9645
1 3	610.7381	251.9481	2.42	0.016	113.6048	1107.871
1.ui	-552.0597	137.3642	-4.02	0.000	-823.1008	-281.0186
age	-2.829917	9.6153	-0.29	0.769	-21.80241	16.14258
_cons	3552.579	269.9778	13.16	0.000	3019.871	4085.288

```
441
442
443
444
445 log close
      name:  <unnamed>
       log:  E:\Documents\STATA\birthweight_assignment2.smcl
  log type:  smcl
 closed on:   1 May 2026, 13:25:29
```